

Modality A: Midterm exam (key)

1. The inner product of two finite-energy waveforms $x(t)$ and $y(t)$ is

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t)y^*(t) dt,$$

with the complex conjugate on the second waveform.

- (a) True ★
(b) False

Solution: Statement 1 of the collection, definition of the inner product.

2. If a waveform $\phi(t)$ has energy E , then the time-scaled waveform $\phi(2t)$ has energy $2E$.

- (a) True
(b) False ★

Solution: Statement 2. The waveform $\phi(at)$ has energy $E/|a|$, so $\phi(2t)$ has energy $\frac{E}{2}$.

3. The waveform $\text{sinc}(t)$ has unit energy and its Fourier transform occupies the band $|f| \leq 1$.

- (a) True
(b) False ★

Solution: Statement 4. The transform of $\text{sinc}(t)$ is a rectangle equal to 1 for $|f| \leq \frac{1}{2}$. The energy is indeed 1.

4. A baseband waveform of bandwidth W , which occupies $|f| \leq W$, allows at most $2W$ symbols per second without inter-symbol interference.

- (a) True ★
(b) False

Solution: Statement 6, Nyquist's criterion: $R_s = 2W$ and $T = \frac{1}{2W}$.

5. A 256-QAM constellation carries 8 bits per symbol, so symbols sent at R_s symbols per second give a bit rate of $8R_s$ bits per second.

- (a) True ★
(b) False

Solution: Statement 7: $R_b = R_s \log_2 M$ with $\log_2 256 = 8$.

6. The energy of the sum of two signals equals the sum of their energies, whatever the two signals are.

- (a) True
(b) False ★

Solution: Statement 8. The energy of a sum is $E_x + E_y + 2 \text{Re}\langle x, y \rangle$, which is the sum of the energies only when $\text{Re}\langle x, y \rangle = 0$.

7. In MATLAB, the expression `2*rand + 1` returns a random number that is uniform on the interval $(1, 2)$.

- (a) True
(b) False ★

Solution: Statement 12. For $a > 0$, $a \cdot \text{rand} + b$ is uniform on $(b, a + b)$, so `2*rand + 1` is uniform on $(1, 3)$.

8. The entropy of a discrete random variable that takes M values is at most $\log_2 M$ bits, and it equals $\log_2 M$ bits when the M values are equiprobable.

- (a) True ★
(b) False

Solution: Statement 13: $H(X) \leq \log_2 |\mathcal{X}|$, with equality for the uniform distribution.

9. Among all real random variables with the same variance, the uniform random variable has the largest differential entropy.

- (a) True
(b) False ★

Solution: Statement 15. The largest differential entropy for a given variance is that of the Gaussian random variable.

10. The capacity of the binary symmetric channel with crossover probability p vanishes at $p = 1$.

- (a) True
(b) False ★

Solution: Statement 17. The capacity $1 - h_2(p)$ equals 1 bit at $p = 0$ and at $p = 1$, and it vanishes at $p = \frac{1}{2}$.

11. The capacity of the complex AWGN channel with average symbol energy E_s and noise variance σ^2 is

$$C = \log_2 \left(1 + \frac{E_s}{\sigma^2} \right)$$

bits per channel use, and it is achieved by a Gaussian input.

- (a) True ★
(b) False

Solution: Statement 19, capacity of the complex AWGN channel.

- 12.** When the symbols are equiprobable, the maximum likelihood detector and the maximum a posteriori detector make the same decisions.
- (a) True *
 - (b) False

Solution: Statement 21: the ML detector coincides with the MAP detector for equiprobable symbols.